

[5253]-181

T.E. (Computer)

THEORY OF COMPUTATION

(2012 Pattern) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data if necessary.

Q1) a) Determine a Regular Expression over the alphabets $\{0, 1\}$ for the following [6]

- i) All strings containing exactly two 0's
- ii) All strings that do not end with 01
- iii) All strings starting with 11

b) Define Pumping Lemma and apply it to prove the following : [6]

$$L = \{0^m 1^n 0^{m+n} \mid m, n \geq 1\} \text{ is not regular.}$$

c) Give the Right & Left linear grammar for the following DFA shown in Fig 1 [8]

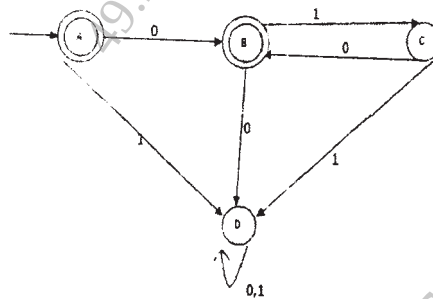


Fig 1

OR

P.T.O.

Q2) a) State Principle of Mathematical Induction and apply it to show that $n^4 - 4n^2$ is divisible by 3 for all $n > 0$. [6]

b) Make use of Arden's theorem to determine the regular expression for the finite automata shown in **Fig 2**. [6]

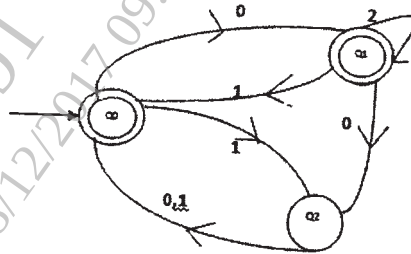


Fig 2

c) Construct a DFA for the following left linear grammar [8]

$$S \rightarrow B1|A0|C0, B \rightarrow B1|1, A \rightarrow A1|B1|C0|0, C \rightarrow A0$$

Q3) a) What is a Turing Machine? Give the formal definition of TM. Design a TM that replaces every occurrence of xyy by yxx . [9]

b) Write short note on : [9]

- Halting Problem of Turing Machine.
- Universal Turing Machine (UTM).
- Recursively Enumerable Languages.

OR

Q4) a) What is a post machine? Give formal definition of Post Machine. Construct a Post Machine for Accepting strings for the language $a^j b^j$. [9]

b) What are the different ways for extension of TM? Explain. Construct a two tape TM to convert an input W into WW^R . [9]

Q5) a) Construct a PDA that accept $L = \{a^n b^n | n \geq 1\}$ through Empty stack [7]

b) Obtain CFG for the PDA given below : [9]

$$\begin{aligned}
 \delta(q_0, 1, z_0) &= \{q_0, xz_0\} & \delta(q_0, 1, x) &= \{q_0, xx\} \\
 \delta(q_0, 0, x) &= \{q_1, x\} & \delta(q_0, \epsilon, z_0) &= \{q_0, \epsilon\} \\
 \delta(q_1, 1, x) &= \{q_1, \epsilon\} & \delta(q_0, 1, z_0) &= \{q_0, z_0\}
 \end{aligned}$$

OR

- Q6)** a) What is PDA? What are the different ways of constructing a PDA? Explain each with example. [8]
- b) What is NPDA? Construct a NPDA for the set of all strings over $\{a,b\}$ with odd length palindrome. [8]
- Q7)** a) What do you mean by NP-Problems? Justify why the Travelling Salesman problem is a NP-Problem. [8]
- b) What do you mean by Polynomial Time Reduction? Explain with suitable example. [8]

OR

- Q8)** a) What is Kruskal's Algorithm? How can we solve this problem using Turing Machine? [8]
- b) What is Clique Problem? Show that it is a NP-Complete problem. [8]
